# Randomized Experiments

BMiSS 2009

Alberto Abadie – Harvard University

# Selection Bias

Recall the selection problem when comparing the mean outcomes for the treated and the untreated:

**Problem**

$$\underbrace{E[Y|D=1] - E[Y|D=0]}_{\textit{Difference in Means}} = E[Y_1|D=1] - E[Y_0|D=0]$$
$$= \underbrace{E[Y_1 - Y_0|D=1]}_{ATET} + \underbrace{\{E[Y_0|D=1] - E[Y_0|D=0]\}}_{BIAS}$$

- Random assignment of units to the treatment forces the selection bias to be zero
- The treatment and control group will tend to be similar along all characteristics, including the potential outcomes under the control condition

# Identification in Randomized Experiments

## Identification Assumption

$(Y_1, Y_0) \perp\!\!\!\perp D$ *(random assignment)*

## Identification Result

*We have that* $E[Y_0|D=1] = E[Y_0|D=0]$ *and therefore*

$$\alpha_{ATET} = E[Y_1 - Y_0|D=1] = E[Y|D=1] - E[Y|D=0]$$

*Also, we have that*

$$\alpha_{ATE} = E[Y_1 - Y_0] = E[Y_1 - Y_0|D=1] = E[Y|D=1] - E[Y|D=0]$$

*As a result,*

$$\underbrace{E[Y|D=1] - E[Y|D=0]}_{\textit{Difference in Means}} = \alpha_{ATE} = \alpha_{ATET}$$

# Identification in Randomized Experiments

## Identification Assumption

*Given random assignment* $(Y_1, Y_0) \perp\!\!\!\perp D$

## Identification Result

$$\begin{aligned} F_{Y_0}(y) &= \Pr(Y_0 \leq y) = \Pr(Y_0 \leq y | D = 0) \\ &= \Pr(Y \leq y | D = 0) \end{aligned}$$

*Similarly,*

$$F_{Y_1}(y) = \Pr(Y \leq y | D = 1).$$

*So effect of the treatment at any quantile,* $Q_\theta(Y_1) - Q_\theta(Y_0)$ *(where* $\theta$ *indexes the quantile and* $F_{Y_d}(Q_\theta(Y_d)) = \theta$*, for* $d = 0, 1$*), is identified.*

- Randomization identifies the whole marginal distributions of $Y_0$ and $Y_1$
- Does not identify the quantiles of the effect: $Q_\theta(Y_1 - Y_0)$ (the difference of quantiles is not the quantile of the difference)

# Estimation in Randomized Experiments

Consider a randomized trial with $N$ individuals.

## Estimand

$\alpha_{ATE} = E[Y_1 - Y_0](= \alpha_{ATET})$

## Estimator

*By the analogy principle we use* $\widehat{\alpha} = \bar{Y}_1 - \bar{Y}_0$ *where*

$$\bar{Y}_1 = \frac{\sum Y_i \cdot D_i}{\sum D_i} = \frac{1}{N_1} \sum_{D_i=1} Y_i \, ;$$

$$\bar{Y}_0 = \frac{\sum Y_i \cdot (1 - D_i)}{\sum (1 - D_i)} = \frac{1}{N_0} \sum_{D_i=0} Y_i$$

*with* $N_1 = \sum_i D_i$ *and* $N_0 = N - N_1$. *It is easy to show that* $\widehat{\alpha}$ *is an unbiased and consistent estimator of* $\alpha_{ATE}$.

# Testing in Large Samples: Two Sample t-Test

From basic statistical theory, we know that

$$\frac{\widehat{\alpha} - \alpha_{ATE}}{\sqrt{\frac{\widehat{\sigma}_1^2}{N_1} + \frac{\widehat{\sigma}_0^2}{N_0}}} \xrightarrow{d} N(0,1),$$

where $\widehat{\sigma}_d^2 = \sum_{D_i=d}(Y_i - \bar{Y}_d)^2/N_d$ $(d \in \{0,1\})$. Let

$$t = \frac{\widehat{\alpha}}{\sqrt{\frac{\widehat{\sigma}_1^2}{N_1} + \frac{\widehat{\sigma}_0^2}{N_0}}}.$$

We reject the null hypothesis $\mathrm{H}_0$: $\alpha_{ATE} = 0$ against the alternative $\mathrm{H}_1$: $\alpha_{ATE} \neq 0$ at the asymptotic 5% significance level if $|t| > 1.96$.

# Testing in Small Samples: Fisher's Exact Test

- Test of differences in means with large $N$:

$$H_0 : E[Y_1] = E[Y_0], \quad H_1 : E[Y_1] \neq E[Y_0]$$

- Fisher's Exact Test with small $N$:

$$H_0 : Y_1 = Y_0, \quad H_1 : Y_1 \neq Y_0 \qquad \text{(sharp null)}$$

- Let $\Omega$ be the set of all possible randomization realizations.
- We only observe the outcomes, $Y_i$, for one realization of the experiment. We calculate $\hat{\alpha} = \bar{Y}_1 - \bar{Y}_0$.
- Under the sharp null hypothesis we can calculate the value that the difference of means would have taken under any other realization, $\hat{\alpha}(\omega)$, for $\omega \in \Omega$.

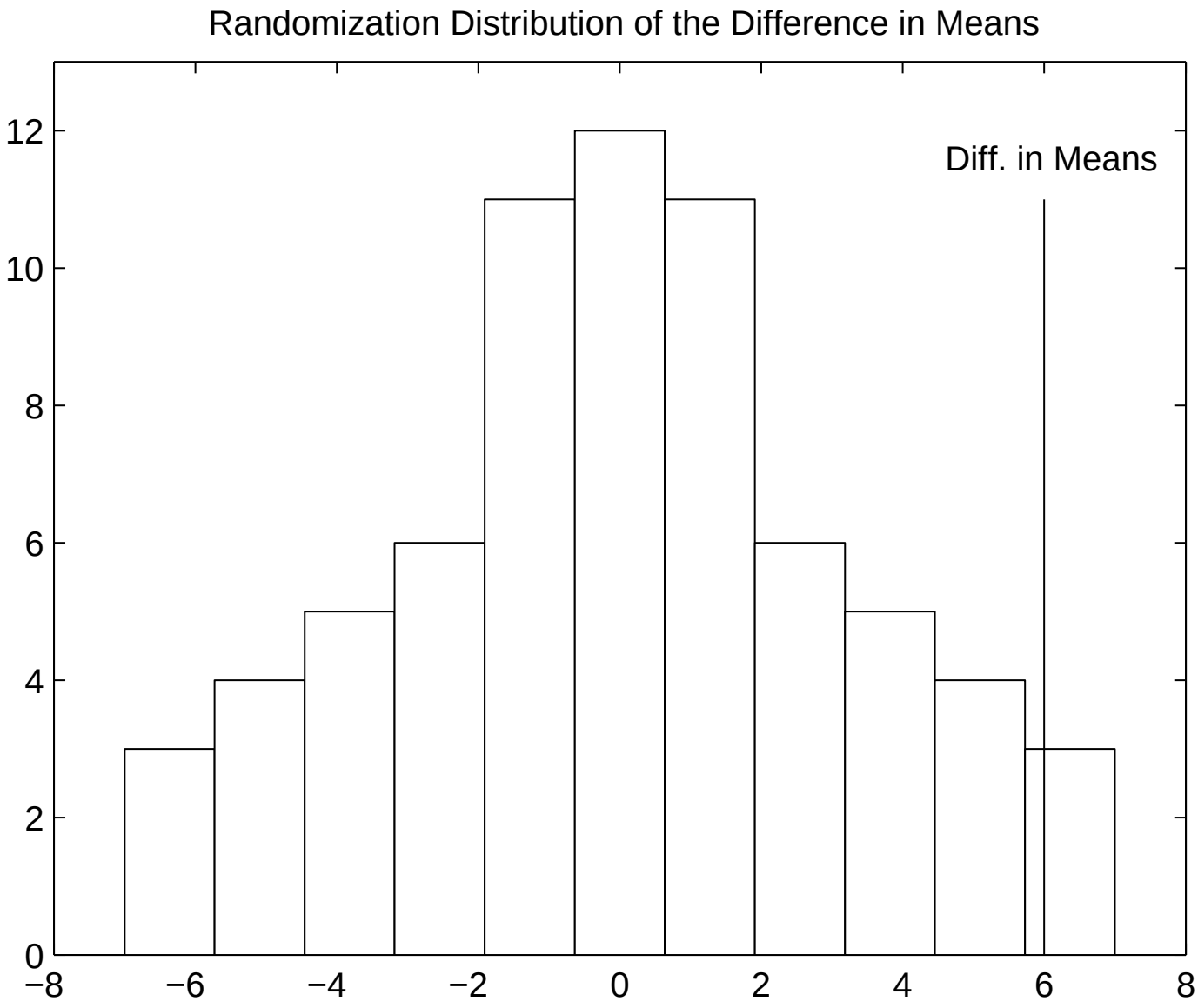
# Testing in Small Samples: Fisher's Exact Test

Suppose that we assign 4 individuals out of 8 to the treatment:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $Y_i$ | 12 | 4 | 6 | 10 | 6 | 0 | 1 | 1 | |
| $D_i$ | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | $\hat{\alpha} = 6$ |
| | | | | | | | | | $\hat{\alpha}(\omega)$ |
| $\omega = 1$ | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 6 |
| $\omega = 2$ | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 4 |
| $\omega = 3$ | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| $\omega = 4$ | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1.5 |
| | | | | $\cdots$ | | | | | |
| $\omega = 70$ | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | -6 |

- The randomization distribution of $\widehat{\alpha}$ (under the sharp null hypothesis) is $\Pr(\widehat{\alpha} \leq z) = \frac{1}{70} \sum_{\omega \in \Omega} 1\{\widehat{\alpha}(\omega) \leq z\}$
- Now, find $\bar{z} = \inf\{z : P(|\widehat{\alpha}| > z) \leq 0.05\}$
- Reject the null hypothesis, $H_0$: $Y_{1i} - Y_{0i} = 0$ for all $i$, against the alternative hypothesis, $H_1$: $Y_{1i} - Y_{0i} \neq 0$ for some $i$, at the 5% significance level if $|\widehat{\alpha}| > \bar{z}$

# Testing in Small Samples: Fisher's Exact Test




$$\Pr(|\hat{\alpha}(\omega)| \geq 6) = 0.0857$$

# Covariate Balance

- Randomization balances observed but also unobserved characteristics between treatment and control group
- Can check random assignment with respect to observed covariates using so called "balance tests" (e.g., t-tests) to see if distributions of the covariates are the same in the treatment and control groups
- $X$ are pre-treatment variables that are measured prior to treatment assignment (i.e., at baseline)

# Experimental Design: Relative sample sizes for fixed $N$

We know that

$$\bar{Y}_1 - \bar{Y}_0 \sim \left(\mu_1 - \mu_0, \frac{\sigma_1^2}{N_1} + \frac{\sigma_0^2}{N_0}\right).$$

**Problem**

*Choose $N_1$ and $N_0$, such that $N_1 + N_0 = N$, to minimize the variance of the estimator of the average treatment effect.*

*The variance of $\bar{Y}_1 - \bar{Y}_0$ is:*

$$var(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma_1^2}{pN} + \frac{\sigma_0^2}{(1-p)N}$$

*where $p = N_1/N$ is the proportion of treated in the sample.*

# Experimental Design: Relative sample sizes for fixed $N$

Find the value $p^*$ that minimizes $\text{var}(\bar{Y}_1 - \bar{Y}_0)$:

$$-\frac{\sigma_1^2}{p^{*2}N} + \frac{\sigma_0^2}{(1-p^*)^2 N} = 0.$$

Therefore:

$$\frac{1-p^*}{p^*} = \frac{\sigma_0}{\sigma_1},$$

and

$$p^* = \frac{\sigma_1}{\sigma_1 + \sigma_0} = \frac{1}{1 + \sigma_0/\sigma_1}$$

A "rule of thumb" for the case $\sigma_1 \approx \sigma_0$ is $p* = 0.5$

For practical reasons it is sometimes better to choose unequal sample sizes (even if $\sigma_1 \approx \sigma_0$)

# Experimental Design: Power calculations to choose $N$

- Recall that for a statistical test:
  - Type I error: Rejecting the null if the null is true
  - Type II error: Not rejecting the null if the null is false
- Size of a test is the probability of type I error. Usually 0.05
- Power of a test is one minus the probability of type II error, i.e. the probability of rejecting the null if the null is false

# Power calculations with equal and known variances

Suppose that $Y_0 \sim (\mu_0, \sigma^2)$ and $Y_1 \sim (\mu_1, \sigma^2)$

Assume also that $p = 0.5$, so $N_0 = N_1 = N/2$. Let $\delta = \mu_1 - \mu_0$.

Then, for the t-statistic of equality of means:

$$\frac{\bar{Y}_1 - \bar{Y}_0 - \delta}{\sqrt{\frac{\sigma_1^2}{N_1} + \frac{\sigma_0^2}{N_0}}} = \frac{\bar{Y}_1 - \bar{Y}_0 - \delta}{\sqrt{\frac{2\sigma^2}{N} + \frac{2\sigma^2}{N}}} = \frac{\bar{Y}_1 - \bar{Y}_0 - \delta}{2\sigma/\sqrt{N}} \sim N(0,1),$$

if the sample is large enough. Therefore:

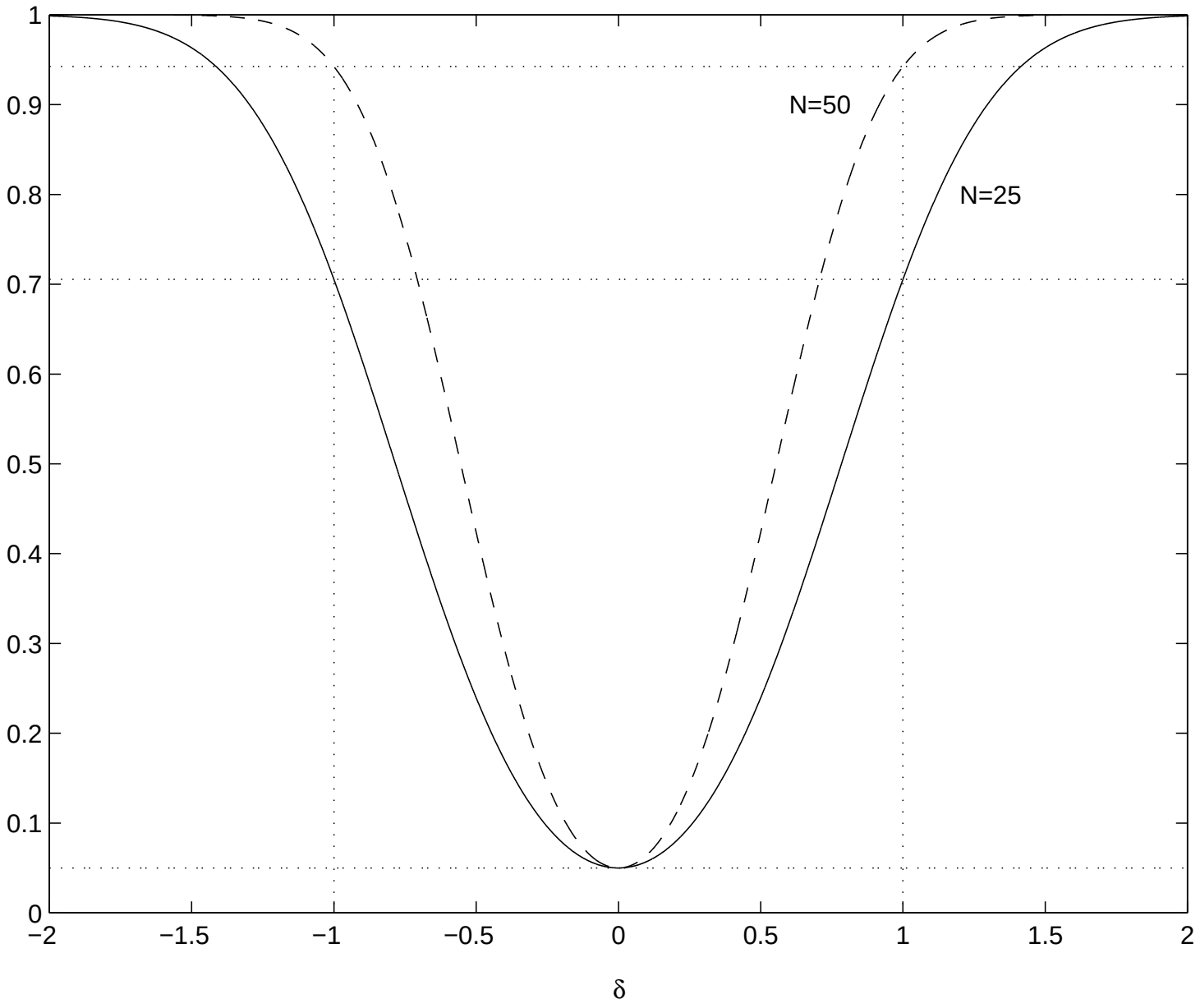
$$t = \frac{\bar{Y}_1 - \bar{Y}_0}{\sqrt{\frac{\sigma_1^2}{N_1} + \frac{\sigma_0^2}{N_0}}} \sim N\left(\frac{\delta\sqrt{N}}{2\sigma}, 1\right)$$

# Power calculations with equal and known variances

The probability of rejecting the null $\mu_1 - \mu_0 = 0$ is:

$$
\begin{aligned}
\Pr\left(|t| > 1.96\right) &= \Pr\left(t < -1.96\right) + \Pr\left(t > 1.96\right) \\
&= \Pr\left(t - \frac{\delta\sqrt{N}}{2\sigma} < -1.96 - \frac{\delta\sqrt{N}}{2\sigma}\right) \\
&+ \Pr\left(t - \frac{\delta\sqrt{N}}{2\sigma} > 1.96 - \frac{\delta\sqrt{N}}{2\sigma}\right) \\
&= \Phi\left(-1.96 - \frac{\delta\sqrt{N}}{2\sigma}\right) + \left(1 - \Phi\left(1.96 - \frac{\delta\sqrt{N}}{2\sigma}\right)\right)
\end{aligned}
$$

# Power functions for $N = 25$, $N = 50$, and $\sigma^2 = 1$



# General formula for the power function ($p \neq 0.5$, $\sigma_0^2 \neq \sigma_1^2$)

$$\Pr\left(\text{reject } \mu_1 - \mu_0 = 0 | \mu_1 - \mu_0 = \delta\right)$$
$$= \Phi\left(-1.96 - \delta \Big/ \sqrt{\frac{\sigma_1^2}{pN} + \frac{\sigma_0^2}{(1-p)N}}\right)$$
$$+ \left(1 - \Phi\left(1.96 - \delta \Big/ \sqrt{\frac{\sigma_1^2}{pN} + \frac{\sigma_0^2}{(1-p)N}}\right)\right).$$

To choose $N$ we need to specify:

1. $\delta$: minimum detectable magnitude of treatment effect
2. Power value (usually 0.80 or higher)
3. $\sigma_1^2$ and $\sigma_0^2$ (usually $\sigma_1^2 = \sigma_0^2$) (e.g., using previous measures)
4. $p$: proportion of observations in the treatment group If $\sigma_1 = \sigma_0$, then the power is maximized by $p = 0.5$

# Threads to the Validity of Randomized Experiments

- Internal validity: can we estimate treatment effect for our particular sample?
  - Fails when there are differences between treated and controls (other than the treatment itself) that affect the outcome and that we cannot control for
- External validity: can we extrapolate our estimates to other populations?
  - Fails when the treatment effect is different outside the evaluation environment

# Most Common Threads to Internal Validity

- Failure of randomization
- Non-compliance with experimental protocol
- Attrition

# Most Common Threads to External Validity

- Non-representative sample
- Non-representative program
  - The treatment differs in actual implementations
  - Scale effects
  - Actual implementations are not randomized (nor full scale)

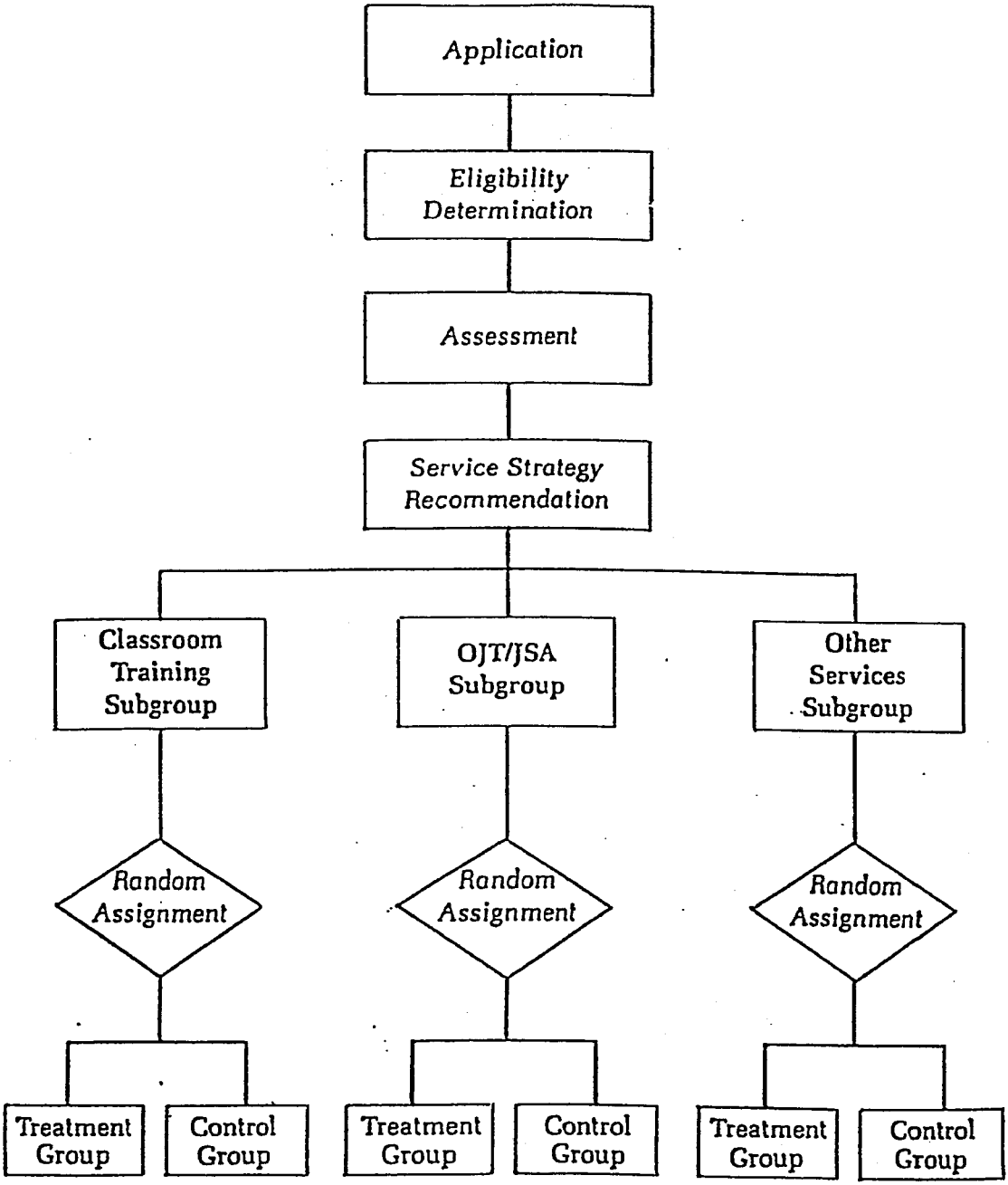
# Example: Job Training Partnership Act (JTPA)

- Largest randomized training evaluation ever undertaken in the U.S.; started in 1983 at 649 sites throughout the country
- Sample: Disadvantaged persons in the labor market (previously unemployed or low earnings)
- D: Assignment to one of three general service strategies
  - classroom training in occupational skills
  - on-the-job training and/or job search assistance
  - other services (eg. probationary employment)
- Y: earnings 30 month following assignment
- X: Characteristics measured before assignment (age, gender, previous earnings, race, etc.)

*Exhibit 5 Impacts on Total 30-Month Earnings: Assignees and Enrollees, by Target Group*

| | *Mean earnings* | | *Impact per assignee* | | |
|---|---|---|---|---|---|
| | *Treatment group (1)* | *Control group (2)* | *In dollars (3)* | *As a percent of (2)* | *Impact per enrollee in dollars* |
| Adult women | $ 13,417 | $ 12,241 | $ 1,176*** | 9.6% | $ 1,837*** |
| Adult men | 19,474 | 18,496 | 978* | 5.3 | 1,599* |
| Female youths | 10,241 | 10,106 | 135 | 1.3 | 210 |
| Male youth non-arrestees | 15,786 | 16,375 | -589 | -3.6 | -868 |
| Male youth arrestees | | | | | |
| Using survey data | 14,633 | 18,842 | -4,209** | -22.3 | -6,804** |
| Using scaled UI data | 14,148 | 14,152 | -4 | 0.0 | -6 |

Sources: Estimates based on First and Second Follow-up Survey responses and earnings data from state unemployment insurance (UI) agencies.
Sample sizes: adult women, 6,102; adult men, 5,102; female youths, 2,657; male youth non-arrestees, 1,704; male youth arrestees, 416.
* Statistically significant at the .10 level, ** at the .05 level, *** at the .01 level (two-tailed test).


Application
Eligibility Determination
Assessment
Service Strategy Recommendation
Classroom Training Subgroup
OJT/JSA Subgroup
Other Services Subgroup
Random Assignment
Random Assignment
Random Assignment
Treatment Group
Control Group
Treatment Group
Control Group
Treatment Group
Control Group

MEANS AND STANDARD DEVIATIONS

| | Entire Sample | Assignment Treatment | Assignment Control | Difference (t-stat.) |
|---|---|---|---|---|
| A. Men | | | | |
| Number of observations | 5,102 | 3,399 | 1,703 | |
| *Treatment* | | | | |
| Training | .42 | .62 | .01 | .61 |
| | [.49] | [.48] | [.11] | (70.34) |
| *Outcome variable* | | | | |
| 30 month earnings | 19,147 | 19,520 | 18,404 | 1,116 |
| | [19,540] | [19,912] | [18,760] | (1.96) |
| *Baseline Characteristics* | | | | |
| Age | 32.91 | 32.85 | 33.04 | -.19 |
| | [9.46] | [9.46] | [9.45] | (-.67) |
| High school or GED | .69 | .69 | .69 | -.00 |
| | [.45] | [.45] | [.45] | (-.12) |
| Married | .35 | .36 | .34 | .02 |
| | [.47] | [.47] | [.46] | (1.64) |
| Black | .25 | .25 | .25 | .00 |
| | [.44] | [.44] | [.44] | (.04) |
| Hispanic | .10 | .10 | .09 | .01 |
| | [.30] | [.30] | [.29] | (.70) |
| Worked less than 13 weeks in past year | .40 | .40 | .40 | .00 |
| | [.47] | [.47] | [.47] | (.56) |
| B. Women | | | | |
| Number of observations | 6,102 | 4,088 | 2,014 | |
| *Treatment* | | | | |
| Training | .45 | .66 | .02 | .64 |
| | [.50] | [.47] | [.13] | (80.24) |
| *Outcome variable* | | | | |
| 30 month earnings | 13,029 | 13,439 | 12,197 | 1,242 |
| | [13,415] | [13,614] | [12,964] | (3.46) |
| *Baseline Characteristics* | | | | |
| Age | 33.33 | 33.33 | 33.35 | -.02 |
| | [9.78] | [9.77] | [9.81] | (-.09) |
| High school or GED | .72 | .73 | .70 | .03 |
| | [.43] | [.43] | [.44] | (2.01) |
| Married | .22 | .22 | .21 | .01 |
| | [.40] | [.40] | [.39] | (1.55) |
| Black | .26 | .27 | .26 | .01 |
| | [.44] | [.44] | [.44] | (.95) |
| Hispanic | .12 | .12 | .12 | -.00 |
| | [.32] | [.32] | [.33] | (-.89) |
| Worked less than 13 weeks in past year | .52 | .52 | .52 | -.00 |
| | [.47] | [.47] | [.47] | (-.08) |
| AFDC | .31 | .30 | .31 | -.01 |
| | [.46] | [.46] | [.46] | (-1.03) |

Exhibit 2.4 DERIVING 30-MONTH EARNINGS SAMPLE FROM FULL EXPERIMENTAL SAMPLE

| | All target groups | Adult women | Adult men | Female youths | Male youth non-arrestees | Male youth arrestees |
|---|---|---|---|---|---|---|
| Full experimental sample | 20,601 | 8,058 | 6,853 | 3,132 | 2,041 | 517 |
| Sample after exogenous deletions for: | | | | | | |
| Extra treatment group members[a] | 20,123 | 7,936 | 6,724 | 3,015 | 1,949 | 499 |
| Late cohorts[b] | 19,019 | 7,497 | 6,303 | 2,864 | 1,871 | 484 |
| Persons in non-UI sites randomly excluded from Second Follow-up survey[c] | 16,347 | 6,191 | 5,223 | 2,712 | 1,755 | 466 |
| Male youth arrestees in non-UI sites[d] | 16,304 | 6,191 | 5,223 | 2,712 | 1,755 | 423 |
| Sample after deletions for missing data: | | | | | | |
| 30-month earnings sample | 15,981 | 6,102 | 5,102 | 2,657 | 1,704 | 416 |
| Potentially nonrandom attrition rate | 2.0% | 1.4% | 2.3% | 2.0% | 2.9% | 1.7% |

a. A total of 473 treatment group members in 5 sites were randomly excluded to ensure a 2/1 treatment/control group ratio in all sites. Also, the 5 sample members under 22 years of age from Oakland, Calif., were deleted because youths were excluded from the experimental design in Oakland.

b. Deleted were all treatment and control group members randomly assigned after December 1988 in Jackson, Miss.; after April 1989 in Butte, Mont., Jersey City, N.J., and Marion, Ohio; and after June 1989 in Omaha, Neb.

c. The "non-UI" sites (where UI earnings data were not available) are Butte, Jersey City, Marion, and Oakland.

d. The remaining sample at this stage has the statistical properties of a randomized experiment.

Exhibit 3.3 SELECTED ECONOMIC CONDITIONS AT 16 STUDY SITES

| Site | Mean unemployment rate, 1987–89 (1) | Mean earnings, 1987 (2) | Percentage employed in manufacturing, mining, or agriculture, 1988 (3) | Annual growth in retail and wholesale earnings, 1989 (4) |
|---|---|---|---|---|
| Fort Wayne, Ind. | 4.7% | $18,700 | 33.3% | −0.1% |
| Coosa Valley, Ga. | 6.5 | 16,000 | 42.8 | 2.1 |
| Corpus Christi, Tex. | 10.2 | 18,700 | 16.8 | −15.5 |
| Jackson, Miss. | 6.1 | 17,600 | 12.8 | −2.4 |
| Providence, R.I. | 3.8 | 17,900 | 28.0 | 9.7 |
| Springfield, Mo. | 5.5 | 15,800 | 19.4 | −1.8 |
| Jersey City, N.J. | 7.3 | 21,400 | 20.9 | 9.9 |
| Marion, Ohio | 7.0 | 18,600 | 37.7 | 1.7 |
| Oakland, Calif. | 6.8 | 23,000 | 14.6 | 3.0 |
| Omaha, Neb. | 4.3 | 18,400 | 11.8 | 1.8 |
| Larimer County, Colo. | 6.5 | 17,800 | 21.2 | −3.1 |
| Heartland, Fla. | 8.5 | 15,700 | 23.8 | −0.3 |
| Northwest Minnesota | 8.0 | 14,100 | 23.0 | 2.4 |
| Butte, Mont. | 6.8 | 16,900 | 9.6 | −5.7 |
| Decatur, Ill. | 9.2 | 21,100 | 27.1 | −1.1 |
| Cedar Rapids, Iowa | 3.6 | 17,900 | 21.9 | −0.5 |
| 16-site average | 6.6 | 18,100 | 22.8 | 0.0 |
| National average, all SDAs | 6.6 | 18,167 | 23.4 | 1.5 |

Source: Unweighted annual averages calculated from JTPA Annual Status Report computer files produced by U.S. Department of Labor.
Note: Missing data for certain measures precluded using same year across columns.

Exhibit 3.6 SELECTED CHARACTERISTICS OF JTPA TITLE II PROGRAMS AT 16 STUDY SITES, PROGRAM YEARS 1987–89

| Site | Mean number of adult and youth terminees[a] (1) | Mean number of weeks enrolled: Adults (2) | Mean number of weeks enrolled: Youths[a] (3) | Mean federal program cost per adult terminee (4) |
|---|---|---|---|---|
| Fort Wayne, Ind. | 1,195 | 16 | 31 | $1,561 |
| Coosa Valley, Ga. | 1,063 | 12 | 15 | 2,481 |
| Corpus Christi, Tex. | 1,049 | 34 | 33 | 2,570 |
| Jackson, Miss. | 1,227 | 8 | 15 | 1,897 |
| Providence, R.I. | 503 | 7 | 5 | 2,841 |
| Springfield, Mo. | 938 | 17 | 17 | 1,898 |
| Jersey City, N.J. | 853 | 16 | 14 | 3,637 |
| Marion, Ohio | 714 | 27 | 26 | 2,199 |
| Oakland, Calif. | 1,396 | 16 | 17 | 2,539 |
| Omaha, Neb. | 1,111 | 11 | 12 | 2,404 |
| Larimer County, Colo. | 354 | 32 | 26 | 1,937 |
| Heartland, Fla. | 1,793 | 15 | 24 | 1,782 |
| Northwest Minnesota | 430 | 29 | 28 | 2,371 |
| Butte, Mont. | 576 | 21 | 19 | 2,665 |
| Decatur, Ill. | 525 | 29 | 25 | 3,039 |
| Cedar Rapids, Iowa | 658 | 31 | 23 | 2,212 |
| 16-site average | 899 | 20 | 21 | 2,377 |
| National average, all SDAs | 1,177 | 20 | 22 | 2,241 |

Source: Unweighted annual averages calculated from JTPA Annual Status Report computer files produced by U.S. Department of Labor.

a. Includes adults and both out-of-school and in-school youths ages 14 to 21. Experimental sample does not include in-school youths or youths under age 16.

# A Final Word about Policy Outcome

After the results of the National JTPA study were released, in 1994, funding for JTPA training for the youth were drastically cut:

Spending on JTPA Programs

| Year | Youth Training Grants | Adult Training Grants |
|---|---|---|
| 1993 | 677 | 1015 |
| 1994 | 609 | 988 |
| 1995 | 127 | 996 |
| 1996 | 127 | 850 |
| 1997 | 127 | 895 |